



Architecture documentation — TASCO Growth Platform

TASCO Insurance × VETC Growth Platform — Architecture

iorta TechNXT

Version 1.0 · 07 October 2026

Contents

Documents

System at a glance

Source map

Conventions used in these documents

TASCO Insurance × VETC motor insurance growth platform (new business, renewals, partners, claims FNOL), built by iorta TechNXT. Runtime: Node.js 22, PostgreSQL 16 (or in-memory for dev/demo), one runtime dependency (pg).

Everything here is written from the source code. Each document cites the files it describes, so a reviewer can check every claim. Where the platform currently uses a **sandbox** adapter rather than a production integration, the document says so. Regulatory statements are flagged "**confirm with TASCO legal**".

Documents

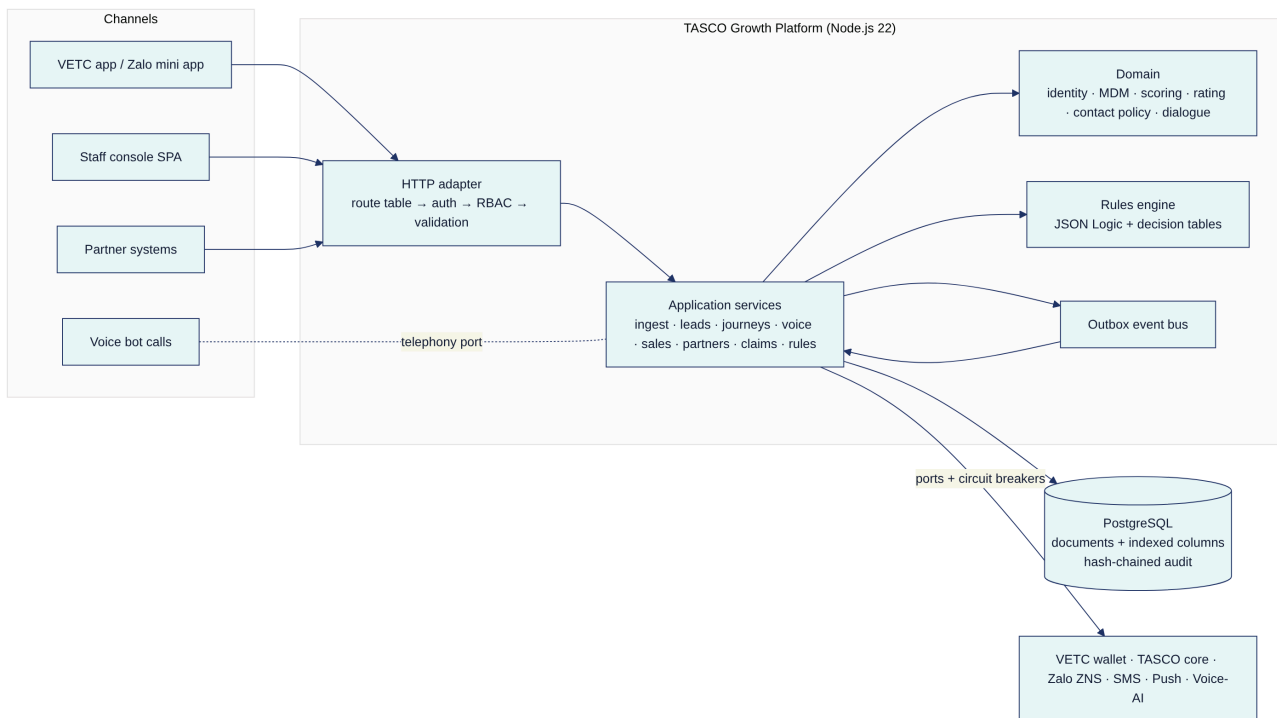
#	Document	Audience	What it answers
1	Solution architecture	CTO, architects, delivery leads	Principles, C4 views, hexagonal layering, key flows, event catalogue, configurability model, scaling to 6M vehicles, roadmap
2	Architecture decision records	Architects, tech leads, auditors	The 12 decisions that shape the platform, with alternatives and consequences
3	Integration architecture	Integration teams (VETC, TASCO core, Zalo, telco, voice-AI vendor, partners)	Every port: contract, protocol, auth, idempotency, resilience, SLA, sandbox vs production
4	Deployment & infrastructure	Platform/SRE, InfoSec, finance	Environments, Kubernetes topology, network zones, Postgres HA, secrets, CI/CD, capacity and cost, observability
5	Data architecture	Data office, DPO, data stewards, BI	ERD, data dictionary, MDM/golden record, data quality, lineage, retention, migration, analytics feed
6	Security architecture	CISO, InfoSec, pen-testers, auditors	STRIDE threat model, OWASP Top 10 and ASVS L2 mapping, RBAC/ABAC, crypto and keys, known gaps
7	AI governance	Compliance, model risk, CX, product	Voice bot principles, script lifecycle, lead-score model risk, LLM policy, monitoring, RACI

ADR index

ADR	Decision	Status
ADR-001	Hexagonal / Clean architecture	Accepted
ADR-002	Node.js 22, minimal dependencies (pg only at runtime), JavaScript + JSDoc	Accepted
ADR-003	JSON Logic rules engine with maker-checker governance	Accepted
ADR-004	PostgreSQL document + indexed-columns pattern; in-memory adapter with the same behaviour	Accepted
ADR-005	Outbox table + SKIP LOCKED relay now, message broker later	Accepted (gap noted)
ADR-006	Field-level AES-256-GCM encryption + HMAC blind index	Accepted
ADR-007	Local JWT + TOTP now; OIDC federation to TASCO IdP as target	Accepted (interim)

ADR	Decision	Status
ADR-008	Hash-chained, append-only audit log	Accepted
ADR-009	Deterministic voice bot dialogue with a governed script; LLM only as an optional classifier behind a port	Accepted
ADR-010	API-first route table that generates OpenAPI	Accepted
ADR-011	Vanilla JS SPA + design tokens (no framework, strict CSP)	Accepted
ADR-012	Containers; Kubernetes as primary target; Railway for pilots	Accepted

System at a glance



Source map

Layer	Path	Contents
Shared kernel	<code>src/shared/</code>	config (12-factor, <code>*_FILE</code> secrets), logger (PII redaction), crypto, validation, resilience, metrics, errors, clock
Rules engine	<code>src/rules/</code>	<code>jsonLogic.js</code> (safe subset), <code>decisionTable.js</code> , <code>validators.js</code> (per-kind validation, copy guard, statutory commission caps)
Domain	<code>src/domain/</code>	<code>identity.js</code> , <code>enrichment.js</code> , <code>leads.js</code> , <code>rating.js</code> , <code>contactPolicy.js</code> , <code>voicebot.js</code>

Layer	Path	Contents
Application	src/application/	One service per use-case area, plus <code>accessPolicy.js</code> (RBAC + ABAC + masking) and <code>rulesService.js</code> (versioning, maker-checker)
Composition root	src/bootstrap/container.js	Chooses adapters, wires services, registers event subscribers
Persistence adapters	src/adapters/persistence/	<code>schema.js</code> (collection registry), <code>codec.js</code> , <code>memoryStore.js</code> , <code>postgresStore.js</code> , <code>auditChain.js</code> , <code>query.js</code>
Messaging adapter	src/adapters/messaging/outboxEventBus.js	Outbox publish / relay / drain
Integration adapters	src/adapters/integrations/	Sandbox VETC wallet, TASCO core, notifications, simulated caller, synthetic VETC source
HTTP adapter	src/adapters/http/	<code>app.js</code> (pipeline), <code>routes.js</code> (73 routes), <code>router.js</code> , <code>openapi.js</code> , <code>security.js</code>
Processes	src/server.js, src/jobs/cli.js	API server with outbox relay and graceful shutdown; batch jobs
Configuration	config/rules/*.json (20 rule kinds), config/security/rbac.json	Business rules seeded as version 1; RBAC
Schema	db/migrations/001_init.sql	19 collection tables + <code>audit_log</code>
Deployment	Dockerfile, docker-compose.yml, deploy/k8s/, railway.json, .github/workflows/ci.yml	See the deployment document

Conventions used in these documents

- **Implemented:** present in the code at the cited path.
- **Sandbox:** the port and the calling code are real; the adapter behind the port simulates the external system (`src/adapters/integrations/*`).
- **Planned / target:** a design commitment that is not in the code yet. Each one names an owner phase in the roadmap.
- **Infra:** a control delivered by the platform (Kubernetes, WAF, managed Postgres, CI) rather than by application code.